

## RESEARCH INTEREST

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- **Operating Systems: Storage Systems and Filesystem**

## EDUCATION

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- **Rutgers University – New Brunswick** New Brunswick, NJ, USA  
*Ph.D. in Computer Sciences, Advisor: Prof. Sudarsun Kannan* *Sept. 2024 – Present*
- **Central South University** Changsha, CN  
*B.E. in Computer Sciences and Technology* *Aug. 2019 – July. 2023*

## PROJECTS

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- **SmartPrefetch (ongoing project in RU system lab)**
  - Extending current prefetching to collaborate with the host cache and device cache.
  - New way of prefetching to leverage remote servers for throughput sensitive tasks.
  - Prefetching to reduce tail latency without affecting throughput.
- **AOS**
  - Developed a self-made OS kernel based on the i386 architecture.
  - Supported multi-threading and loading user programs.
  - Implemented a simple ext2-like file system.
  - Included basic I/O and memory management.
  - Created a simple shell for user interaction.

## HONOR & AWARD

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- The 2020 ICPC Asia Shenyang Regional Contest, Shenyang, Silver Medal *July. 2021*
- The 2021 ICPC Asia Shanghai Regional Contest, Shanghai, Bronze Medal *Nov. 2021*
- The 2021 ICPC Asia Shenyang Regional Contest, Shenyang, Bronze Medal *Nov. 2021*
- The 2021 CCPC - National Invitational Contest(Hunan), Xiangtan, Silver Medal *Jun. 2021*
- The 2021 Baidu Astar Programming Contest Rank 245/23051 *Aug. 2021*

## EXPERIENCE

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- **Rutgers University** New Brunswick, NJ, USA  
*Teaching Assistant* *Sept. 2024 - Present*
  - Teaching assistant for CS344, undergrad level algorithms course.
- **ByteDance** *data-arch-IES* Shanghai, CN  
*Software Engineer Intern* *Feb 2022 - July 2022*
  - Processed real-time data from Douyin and TikTok utilizing the VCP framework, an advanced and customized version of Apache Flink.
  - Designed and implemented robust data pipelines to ensure efficient and reliable data flow.
  - Conducted performance tuning and optimization of workflows to enhance processing speed and resource utilization.
  - Scaled streaming solutions to accommodate increasing data volumes and ensure seamless performance.
  - Developed and integrated comprehensive performance monitoring tools to track and analyze system metrics, ensuring optimal operation and quick issue resolution.
- **Tencent** *IEG-START Cloud Game* Shenzhen, CN  
*Game Development Engineer Intern* *July 2022 - Aug 2022*
  - Implemented and optimized the rendering model for tree leaves and branches
  - Enhanced the build system for projects
  - Conducted testing and debugging to ensure smooth gameplay.

- Collaborated with the development team for seamless integration of enhancements.

- **CSUVAG**

*Research Assistant Intern*, Advisor: Prof. Jiazhi Xia

Changsha, CN

*Aug 2022 - May 2023*

- Used and implemented multiple deep metric learning and manifold learning techniques to reduce the dimensions and generate projections of high-dimension data.
- Built a visualization system to generate the low-dimension projection of folding and unfolding of high-dimension data.

## PROGRAMMING SKILLS

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- **Languages:** C, C++, Python, Java, Verilog
- **Technologies:** Linux, Git, PyTorch